

Linux VM Migration

How hyper2kvm Migrates Linux from VMware, Hyper-V, AWS, Azure & GCP to KVM

Automated initramfs rebuild with VirtIO drivers, fstab UUID repair, GRUB reconfiguration, SELinux relabeling, network cleanup, and boot verification. Supports RHEL, Ubuntu, Debian, SUSE, Fedora, CentOS, Rocky, Alma, Oracle Linux, Photon OS, and more — across BIOS, UEFI, LVM, XFS, ext4, and Btrfs.

Auto initramfs | fstab Fix | GRUB Repair | SELinux Safe | 15+ Distros | Apache 2.0

The Linux Boot Problem

Why Linux VMs fail to boot after migration — and how hyper2kvm prevents every failure.

Without initramfs rebuild and fstab repair, most Linux VMs fail to boot on KVM

Linux loads storage drivers from the initramfs at boot time. VMware VMs use `vmw_pvscsi` or `mptscsi`. Hyper-V uses `hv_storvsc`. AWS uses `xen-blkfront` or `nvme`. KVM needs `virtio_blk` / `virtio_scsi`. If VirtIO drivers aren't in the initramfs, the kernel can't mount the root filesystem and drops to an emergency shell — or panics. Meanwhile, fstab entries referencing `/dev/sda` or `/dev/xvda` break because KVM presents disks as `/dev/vda`.

Every Failure Mode — Automatically Fixed

Failure	Symptom	Root Cause	hyper2kvm Automatic Fix
Kernel panic	VFS: Unable to mount root fs	Missing virtio_blk/virtio_scsi in initramfs	Rebuilds initramfs with all VirtIO modules injected
Root mount failure	Dropped to emergency shell	fstab uses /dev/sda1 — KVM has /dev/vda1	Converts all fstab entries to UUID-based mounts
GRUB boot failure	grub rescue> prompt	root= parameter points to wrong device	Updates GRUB config with correct root= device/UUID
No network	No interfaces up	VMware vmxnet3/Hyper-V hv_netvsc config gone	Removes old NIC configs, ensures virtio_net loads
SELinux denial storm	Services fail, permission denied	File contexts wrong after disk migration	Triggers full SELinux autorelabel on first boot
VMware Tools errors	Service failures in journal	open-vm-tools can't find hypervisor	Removes open-vm-tools packages and services
Hyper-V daemons crash	hv_kvp_daemon errors	Hyper-V IC daemons without Hyper-V	Removes hv_utils, hv_kvp_daemon, hv_vss_daemon
Cloud-init re-provision	Hostname/network reset	cloud-init runs on new "instance"	Disables or reconfigures cloud-init for KVM
LVM not found	Volume group missing	initramfs lacks dm-mod or LVM tools	Ensures LVM modules + tools included in initramfs
Swap mount failure	Boot delayed or fails	Swap partition device name changed	Updates swap entry in fstab to UUID

All fixes applied offline — hyper2kvm mounts the disk via qemu-nbd, modifies the filesystem directly (fstab, GRUB, initramfs, network configs), then unmounts. The VM boots clean on KVM the first time.

Supported Distributions

Every major enterprise and community Linux distribution — tested and verified.

Distribution	Versions	Init System	Boot Mode	Filesystems	Status	initramfs Tool
RHEL	7, 8, 9, 10	systemd	UEFI, BIOS	XFS, ext4, LVM	Tested	dracut
Rocky Linux	8, 9	systemd	UEFI, BIOS	XFS, ext4, LVM	Tested	dracut
AlmaLinux	8, 9	systemd	UEFI, BIOS	XFS, ext4, LVM	Tested	dracut
CentOS Stream	8, 9	systemd	UEFI, BIOS	XFS, ext4, LVM	Tested	dracut
Oracle Linux	7, 8, 9	systemd	UEFI, BIOS	XFS, ext4, LVM	Tested	dracut
Fedora	38, 39, 40, 41	systemd	UEFI, BIOS	XFS, Btrfs, ext4	Tested	dracut
Ubuntu	20.04, 22.04, 24.04	systemd	UEFI, BIOS	ext4, ZFS, LVM	Tested	update-initramfs
Debian	11, 12	systemd	UEFI, BIOS	ext4, XFS, LVM	Tested	update-initramfs
SUSE SLES	12 SP5, 15 SP5/SP6	systemd	UEFI, BIOS	Btrfs, XFS, ext4	Tested	dracut
openSUSE	Leap 15.5/15.6, TW	systemd	UEFI, BIOS	Btrfs, XFS	Tested	dracut
VMware Photon OS	3, 4, 5	systemd	UEFI, BIOS	ext4	Tested	dracut
Amazon Linux	2, 2023	systemd	UEFI, BIOS	XFS, ext4	Tested	dracut
CentOS	7 (legacy)	systemd	BIOS	XFS, ext4, LVM	Tested	dracut

Filesystem & Volume Manager Support

Filesystems

- **ext4** — most common, full support
- **XFS** — RHEL default, UUID-based mount
- **Btrfs** — SUSE/Fedora, subvolume-aware
- **ZFS** — Ubuntu, pool import handled

Volume Management

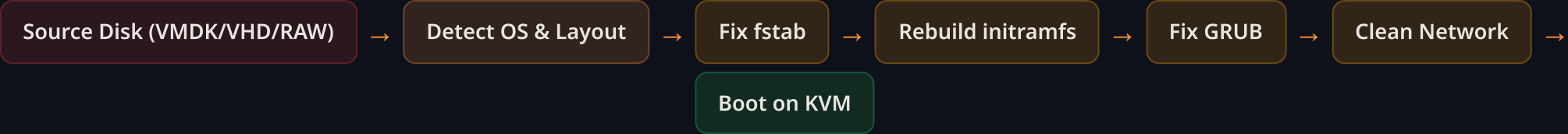
- **LVM2** — VG/LV detection, module injection
- **mdadm RAID** — software RAID arrays
- **Plain partitions** — GPT and MBR
- **LUKS encryption** — encrypted root support

Boot Loaders

- **GRUB2** — primary, config regeneration
- **GRUB Legacy** — CentOS 6/RHEL 6
- **systemd-boot** — modern UEFI systems
- **UEFI + BIOS** — both modes supported

The Fix Pipeline

What hyper2kvm does to every Linux VM — automatically, offline, before first boot.



#	Fix	What It Does	Technical Detail
1	OS Detection	Identifies distro, version, init system, boot mode, partitions	Parses /etc/os-release, EFI partition presence, partition table (GPT/MBR), LVM VGs
2	fstab UUID Repair	Converts device-name mounts (/dev/sda1) to UUID-based	Reads blkid for each partition, replaces /dev/sdX, /dev/xvdX, /dev/hdX with UUID=xxxx in /etc/fstab
3	initramfs Rebuild	Injects VirtIO drivers: virtio_blk, virtio_scsi, virtio_net, virtio_pci	Chroot into guest, run dracut (RHEL/Fedora/SUSE) or update-initramfs (Debian/Ubuntu) with --add-drivers
4	GRUB Update	Fixes root= kernel parameter for KVM disk device	Updates /etc/default/grub GRUB_CMDLINE_LINUX, regenerates grub.cfg via grub2-mkconfig
5	Network Cleanup	Removes old NIC configs (vmxnet3/hv_netvsc device names)	Deletes /etc/sysconfig/network-scripts/ifcfg-*, /etc/netplan/*.yaml with hardware-specific MACs, udev rules in 70-persistent-net.rules
6	VMware Tools Removal	Removes open-vm-tools packages and services	Chroot: rpm -e open-vm-tools or apt remove open-vm-tools, disable vmtoolsd.service
7	Hyper-V IC Removal	Removes Hyper-V integration daemons	Disable hv_kvp_daemon, hv_vss_daemon, hv_fccopy_daemon services
8	Cloud Agent Cleanup	Disables AWS/Azure/GCP agents that fail without cloud metadata	Disable cloud-init reprovisioning, waagent, SSM agent; preserve hostname
9	SELinux Relabel	Triggers full filesystem relabel on next boot	Creates /.autorelabel — SELinux restorecon runs on all files at first boot
10	Serial Console	Enables ttyS0 console for headless access	Adds console=ttyS0,115200n8 to GRUB_CMDLINE_LINUX, enables serial-getty@ttyS0

initramfs Rebuild — Distro-Specific Commands

```
# RHEL / Rocky / Alma / Fedora / CentOS / Oracle Linux / Photon / SUSE (dracut-based):
chroot /mnt/guest dracut --force --add-drivers "virtio_blk virtio_scsi virtio_net virtio_pci" /boot/initramfs-$(uname -r).img $(uname -r)

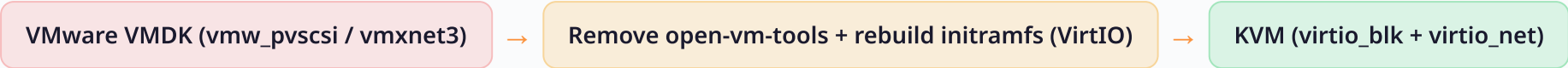
# Ubuntu / Debian (initramfs-tools-based):
echo -e "virtio_blk\nvirtio_scsi\nvirtio_net\nvirtio_pci" >> /mnt/guest/etc/initramfs-tools/modules
chroot /mnt/guest update-initramfs -u -k all

# hyper2kvm handles all of this automatically – detects distro, selects correct method, runs in chroot
```

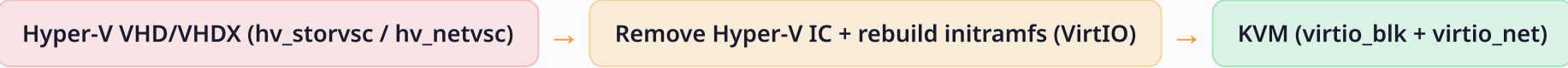
Every Source Platform

hyper2kvm migrates Linux VMs from any hypervisor or cloud with source-specific fixes.

VMware vSphere → KVM



Hyper-V → KVM



AWS EC2 → KVM



Azure → KVM



Source-Specific Fix Matrix

Fix Applied	VMware	Hyper-V	AWS EC2	Azure	GCP	OpenStack
initramfs + VirtIO	✓	✓	✓	✓	✓	✓
fstab UUID repair	✓	✓	✓	✓	✓	✓
GRUB config fix	✓	✓	✓	✓	✓	✓
Network cleanup	✓	✓	✓	✓	✓	✓
SELinux relabel	✓	✓	✓	✓	✓	✓
Remove VMware Tools	✓	N/A	N/A	N/A	N/A	N/A
Remove Hyper-V IC	N/A	✓	N/A	✓	N/A	N/A

Disable cloud-init	N/A	N/A	✓	✓	✓	✓
Remove cloud agents	N/A	N/A	✓ (SSM)	✓ (waagent)	✓ (guest-agent)	N/A
Serial console	✓	✓	✓	✓	✓	✓

Enterprise Linux Workloads

Critical applications migrated safely with hyper2kvm.

PostgreSQL / MySQL / MariaDB

- Data directories migrate with disk — no dump/restore
- InnoDB/WAL files intact on target
- VirtIO SCSI provides comparable I/O to PVSCSI
- Tablespace on separate partitions handled via fstab fix
- Replication resumes after boot (update source IP)
- **Tip:** Stop DB before export for consistency, or use snapshot

Docker / Podman Hosts

- Container images in /var/lib/docker migrate with disk
- Overlay2/fuse-overlayfs storage intact
- Docker daemon starts on boot, containers resume
- Podman rootless containers preserved
- Container registry data (Harbor, GitLab) migrates with volume
- **Tip:** docker-compose stacks auto-restart via systemd

Kubernetes Nodes

- kubelet, containerd, etcd data migrates with disk
- K8s control plane: migrate workers first, then masters
- etcd snapshot recommended before control plane migration
- CNI plugins (Calico, Cilium) need network reconfiguration
- PersistentVolumes on local disks travel with the VM
- **Tip:** Cordon + drain node before migration

Web Servers (Apache / Nginx)

- Simplest migration — standard pipeline, no special handling
- Virtual hosts, SSL certs, .htaccess all preserved
- Let's Encrypt certs in /etc/letsencrypt intact
- PHP-FPM, Python WSGI, Node.js apps unchanged
- Load balancer health checks resume automatically
- **Tip:** Verify DNS and IP bindings post-migration

NFS / Samba File Servers

- Exported filesystems migrate with disk
- NFS exports in /etc/exports preserved
- Samba shares in smb.conf intact
- POSIX ACLs and extended attributes preserved on ext4/XFS
- Clients reconnect after DNS/IP update
- **Tip:** Update /etc/exports for new network if IP changes

Monitoring (Prometheus / Grafana / ELK)

- Prometheus TSDB data in /var/lib/prometheus intact
- Grafana dashboards in SQLite/PostgreSQL preserved
- Elasticsearch indices on data volumes travel with disk
- Alertmanager configs and templates preserved
- Large data volumes: ensure target has sufficient I/O
- **Tip:** VirtIO SCSI + NVMe backend for heavy I/O

GitLab / Jenkins CI

- Git repositories, CI data, artifacts migrate with disk
- GitLab PostgreSQL + Redis data intact
- Jenkins workspace and job history preserved
- SSH keys and deploy keys in /home/*.ssh intact
- Runner registrations: re-register with new IP
- **Tip:** GitLab backup before migration as safety net

FreeIPA / LDAP / Kerberos

- LDAP directory (389-ds) data migrates with disk
- Kerberos keytabs and principal database preserved
- CA certificates in /etc/pki intact
- DNS zones (if integrated) travel with LDAP
- Replication: migrate replicas first, verify sync
- **Tip:** Test kinit + ldapsearch post-migration

Configuration & Deployment

YAML-driven migration with dual deployment targets.

VMware Linux VM → KVM

```
# rhel9-webserver.yaml
cmd: vsphere
vcenter: vcenter.corp.com
vs_action: export_vm
vs_vm: rhel9-web-01
output_dir: ./converted
out_format: qcow2
flatten: true
compress: true
regen_initramfs: true
emit_domain_xml: true
vm_name: rhel9-web-01
memory: 4096
vcpus: 2
machine: q35
uefi: true
boot_test: true
```

AWS Linux VM → KubeVirt

```
# aws-postgres.yaml
cmd: local
raw: ./aws-export/postgres-db.raw
output_dir: ./converted
out_format: qcow2
regen_initramfs: true
deploy_k8s: true
k8s_namespace: databases
vm_name: postgres-primary
memory: 16384
vcpus: 8
boot_test: true
# hypersdk exported the RAW from
# AWS EBS snapshot earlier
```

Deployment Targets

KVM / libvirt

- `emit_domain_xml: true`
- Generates optimized libvirt domain XML
- Q35 machine type with PCIe for modern guests
- VirtIO disk, network, balloon, RNG
- UEFI (OVMF) or BIOS boot
- Auto virsh define + optional boot test

KubeVirt / Kubernetes

- `deploy_k8s: true`
- Generates VirtualMachine manifest
- PVC with disk image upload via CDI
- Works on K3s, vanilla K8s, OpenShift
- GitOps-ready YAML for fleet deploy
- kubectl apply to any cluster

Dual Target

- `emit_domain_xml: true`
- `deploy_k8s: true`
- Both outputs from one conversion
- KVM for traditional hosts
- KubeVirt for K8s clusters
- Choose per-site at deploy time

Batch Migration — 100 Linux VMs

#	Step	What hyper2kvm Does	Result
1	Pre-flight	<code>h2kvmctl doctor</code> — checks disk space, qemu-nbd, libvirt, KubeVirt	Green light before starting
2	Discover	zkvm TUI: browse vCenter, see OS, CPU, RAM, disk per VM	Select VMs, batch queue
3	Export	govc export with streaming progress + ETA	VMDK/OVA files downloaded
4	Convert	VMDK → QCOW2, flatten snapshots, compress	Optimized QCOW2 images
5	Fix	fstab, initramfs, GRUB, network, SELinux — per distro	Boot-ready on KVM
6	Deploy	Generate XML/YAML, virsh define or kubectl apply	VMs defined on target
7	Verify	Automated boot test, service health check	Confirmed working
8	Notify	Webhook to Slack/Teams with pass/fail per VM	Team informed

Linux Migration Tool Comparison

How hyper2kvm compares to other tools for Linux VM migration.

Capability	hyper2kvm	virt-v2v	Red Hat MTV	AWS MGN	Manual
initramfs rebuild + VirtIO	✓ Auto (all distros)	✓ Auto	Limited	N/A	Manual chroot
fstab UUID repair	✓ Auto	✓	Basic	N/A	Manual edit
GRUB config fix	✓ Auto	✓	Basic	N/A	Manual
SELinux relabel	✓ Auto	✓	Partial	N/A	Manual
Network cleanup	✓ Auto	✓	Basic	N/A	Manual
Cloud agent removal	✓ (AWS/Azure/GCP)	✗	✗	N/A	Manual
Boot verification	✓ Auto	✗	✗	✗	Manual
Deploy to KVM/libvirt	✓ Auto XML	✓	✗	✗	Manual
Deploy to KubeVirt	✓ Auto YAML	✗	✓ (OCP only)	✗	Complex
Interactive TUI	✓ zkvm	✗	Web UI	✗	✗
Batch migration	✓	✗	✓	✓	✗
Source: VMware	✓	✓	✓	✓	Manual
Source: Hyper-V	✓	✓	Partial	✓	Manual
Source: AWS/Azure/GCP	✓ (10 clouds)	✗	✗	AWS only	Complex
No libguestfs needed	✓ VMCraft	Required	Required	N/A	N/A
15+ distros supported	✓	✓	RHEL-focused	Amazon Linux	Varies
Cost	\$0	\$0	\$50K+ (OCP)	Free/90d	\$0

hyper2kvm advantage: The only tool that migrates Linux from VMware + Hyper-V + 10 clouds, supports 15+ distros, deploys to both KVM and KubeVirt, includes automated boot verification, needs no libguestfs, and costs \$0.

Real-World Scenarios

Enterprise Linux migration projects powered by hyper2kvm.

VMware Exit — 200 RHEL/Ubuntu Servers

- **Source:** vSphere 7, mix of RHEL 8/9, Ubuntu 22.04
- **Workloads:** Web, DB, middleware, monitoring, CI/CD
- **Target:** KVM on bare-metal RHEL 9 hosts
- **Process:** zkvm TUI batch export, auto initramfs + fstab + GRUB fix per distro, boot verification
- **Timeline:** 6 weeks (30-40 VMs/week)
- **Savings:** \$450K/yr VMware licensing eliminated

AWS Repatriation — 50 Linux VMs

- **Source:** AWS EC2 (Amazon Linux 2, Ubuntu, RHEL)
- **Workloads:** PostgreSQL, Django apps, Elasticsearch
- **Target:** On-prem KVM (data sovereignty)
- **Process:** hypersdk exports EBS snapshots, hyper2kvm converts RAW → QCOW2, removes cloud-init/SSM, rebuilds initramfs, deploys to libvirt
- **Timeline:** 3 weeks
- **Savings:** \$600K/yr in AWS compute + egress

Hyper-V to KubeVirt — 30 Linux VMs

- **Source:** Hyper-V 2019, Debian 11/12, Ubuntu 20.04
- **Workloads:** Docker hosts, GitLab, Jenkins, Nginx
- **Target:** K3s + KubeVirt (modernization)
- **Process:** VHD export, remove Hyper-V IC, rebuild initramfs, generate KubeVirt manifests
- **Timeline:** 2 weeks
- **Savings:** \$80K/yr Hyper-V + Windows Server licensing

Multi-Cloud Consolidation — 80 VMs

- **Source:** 30 on AWS, 20 on Azure, 15 on GCP, 15 on VMware
- **Workloads:** RHEL, Ubuntu, SUSE, Photon OS — everything
- **Target:** Centralized KVM cluster + edge K3s sites
- **Process:** hypersdk exports from all 4 sources, hyper2kvm applies source-specific fixes per distro, dual-target deploy
- **Timeline:** 5 weeks
- **Savings:** \$1.2M/yr consolidated infrastructure

Linux Migration Made Automatic

15+ distributions. 10 source platforms. Every guest fix automated.
initramfs rebuilt. fstab repaired. GRUB fixed. SELinux relabeled. Boot verified.
hyper2kvm: every Linux VM boots clean on KVM, first time, every time.

